

PRACTICAL-4

QUESTION 1 – CROSS JOIN

```
mysql> SELECT *  
      -> FROM class  
      -> CROSS JOIN class_info;  
+-----+-----+-----+-----+  
| id  | name | id  | address |  
+-----+-----+-----+-----+  
| 4   | alex | 1   | DELHI   |  
| 2   | adam | 1   | DELHI   |  
| 1   | abhi | 1   | DELHI   |  
| 4   | alex | 2   | MUMBAI  |  
| 2   | adam | 2   | MUMBAI  |  
| 1   | abhi | 2   | MUMBAI  |  
| 4   | alex | 3   | CHENNAI |  
| 2   | adam | 3   | CHENNAI |  
| 1   | abhi | 3   | CHENNAI |  
+-----+-----+-----+-----+  
9 rows in set (0.00 sec)
```

QUESTION 2 - INNER JOIN

```
mysql> SELECT *  
      -> FROM class  
      -> INNER JOIN class_info  
      -> ON class.id = class_info.id;  
+-----+-----+-----+-----+  
| id  | name | id  | address |  
+-----+-----+-----+-----+  
| 1   | abhi | 1   | DELHI   |  
| 2   | adam | 2   | MUMBAI  |  
| 3   | alex | 3   | CHENNAI |  
+-----+-----+-----+-----+  
3 rows in set (0.00 sec)
```

QUESTION 3 - INNER JOIN

```
mysql>
mysql> SELECT class.name,
->         class_info.address
-> FROM class
-> INNER JOIN class_info
-> ON class.id=class_info.id;
+-----+-----+
| name | address |
+-----+-----+
| abhi | DELHI   |
| adam | MUMBAI  |
| alex | CHENNAI |
+-----+-----+
3 rows in set (0.00 sec)
```

QUESTION 4 - NATURAL JOIN

```
mysql> SELECT *
-> FROM class
-> NATURAL JOIN class_info;
+-----+-----+-----+
| id  | name | address |
+-----+-----+-----+
| 1   | abhi | DELHI   |
| 2   | adam | MUMBAI  |
| 3   | alex | CHENNAI |
+-----+-----+-----+
3 rows in set (0.00 sec)
```

QUESTION 5 - LEFT JOIN

```
mysql>
mysql> SELECT *
      -> FROM class
      -> LEFT OUTER JOIN class_info
      -> ON class.id=class_info.id;
+-----+-----+-----+-----+
| id    | name  | id    | address |
+-----+-----+-----+-----+
| 1     | abhi  | 1     | DELHI   |
| 2     | adam  | 2     | MUMBAI  |
| 3     | alex  | 3     | CHENNAI |
| 4     | anu   | NULL  | NULL    |
| 5     | ashish| NULL  | NULL    |
+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

QUESTION 6 - LEFT JOIN

```
mysql> SELECT *
      -> FROM class
      -> LEFT JOIN class_info
      -> ON class.id=class_info.id
      -> WHERE class_info.id IS NULL;
+-----+-----+-----+-----+
| id    | name  | id    | address |
+-----+-----+-----+-----+
| 4     | anu   | NULL  | NULL    |
| 5     | ashish| NULL  | NULL    |
+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

QUESTION 7 - RIGHT JOIN

```
mysql> SELECT *  
      -> FROM class  
      -> RIGHT OUTER JOIN class_info  
      -> ON class.id=class_info.id;  
+-----+-----+-----+-----+  
| id    | name  | id    | address |  
+-----+-----+-----+-----+  
| 1     | abhi  | 1     | DELHI   |  
| 2     | adam  | 2     | MUMBAI  |  
| 3     | alex  | 3     | CHENNAI |  
| NULL  | NULL  | 7     | NOIDA   |  
| NULL  | NULL  | 8     | PANIPAT |  
+-----+-----+-----+-----+  
5 rows in set (0.00 sec)
```

QUESTION 8 - RIGHT JOIN

```
mysql> SELECT *  
      -> FROM class  
      -> RIGHT JOIN class_info  
      -> ON class.id=class_info.id  
      -> WHERE class.id IS NULL;  
+-----+-----+-----+-----+  
| id    | name  | id    | address |  
+-----+-----+-----+-----+  
| NULL  | NULL  | 7     | NOIDA   |  
| NULL  | NULL  | 8     | PANIPAT |  
+-----+-----+-----+-----+  
2 rows in set (0.00 sec)
```

QUESTION 9 - FULL OUTER JOIN

```
mysql> SELECT *  
-> FROM class  
-> LEFT JOIN class_info  
-> ON class.id = class_info.id  
->  
-> UNION  
->  
-> SELECT *  
-> FROM class  
-> RIGHT JOIN class_info  
-> ON class.id = class_info.id;
```

id	name	id	address
1	abhi	1	DELHI
2	adam	2	MUMBAI
3	alex	3	CHENNAI
4	anu	NULL	NULL
5	ashish	NULL	NULL
NULL	NULL	7	NOIDA
NULL	NULL	8	PANIPAT

7 rows in set (0.01 sec)

QUESTION 10 - FULL OUTER JOIN

```
mysql> SELECT *  
-> FROM class  
-> LEFT JOIN class_info  
-> ON class.id = class_info.id  
-> WHERE class_info.id IS NULL  
->  
-> UNION  
->  
-> SELECT *  
-> FROM class  
-> RIGHT JOIN class_info  
-> ON class.id = class_info.id  
-> WHERE class.id IS NULL;
```

id	name	id	address
4	anu	NULL	NULL
5	ashish	NULL	NULL
NULL	NULL	7	NOIDA
NULL	NULL	8	PANIPAT

4 rows in set (0.00 sec)

QUESTION 11 - UNION

```
mysql> SELECT * FROM first_table  
-> UNION  
-> SELECT * FROM second_table;
```

id	name
1	abhi
2	adam
3	chester

3 rows in set (0.00 sec)

QUESTION 12 - UNION

```
mysql> SELECT name FROM first_table
-> UNION
-> SELECT name FROM second_table;
+-----+
| name  |
+-----+
| abhi  |
| adam  |
| chester |
+-----+
3 rows in set (0.00 sec)
```

QUESTION 13 - UNION ALL

```
mysql> SELECT * FROM first_table
-> UNION ALL
-> SELECT * FROM second_table;
+-----+-----+
| id  | name  |
+-----+-----+
| 1   | abhi  |
| 2   | adam  |
| 2   | adam  |
| 3   | chester |
+-----+-----+
4 rows in set (0.00 sec)
```

QUESTION 14 - UNION ALL

```
mysql> SELECT COUNT(*)
-> FROM
-> (
-> SELECT * FROM first_table
-> UNION ALL
-> SELECT * FROM second_table
-> ) A;
+-----+
| COUNT(*) |
+-----+
| 4        |
+-----+
1 row in set (0.00 sec)
```

QUESTION 15 - INTERSECT

```
mysql> SELECT * FROM first_table
-> INTERSECT
-> SELECT * FROM second_table;
+-----+-----+
| id    | name |
+-----+-----+
|      2 | adam |
+-----+-----+
1 row in set (0.00 sec)
```

QUESTION 16 - INTERSECT

```
mysql> SELECT name FROM first_table
-> INTERSECT
-> SELECT name FROM second_table;
+-----+
| name |
+-----+
| adam |
+-----+
1 row in set (0.00 sec)
```

QUESTION 17 - MINUS

```
mysql> SELECT *
-> FROM first_table f
-> WHERE NOT EXISTS (
->     SELECT 1
->     FROM second_table s
->     WHERE f.id = s.id
->     AND f.name = s.name
-> );
+-----+-----+
| id    | name |
+-----+-----+
|      1 | abhi |
+-----+-----+
1 row in set (0.01 sec)
```

QUESTION 18 - MINUS

```
mysql> SELECT name
-> FROM first_table f
-> WHERE NOT EXISTS (
->     SELECT 1
->     FROM second_table s
->     WHERE f.name = s.name
-> );
+-----+
| name |
+-----+
| abhi |
+-----+
1 row in set (0.00 sec)
```

QUESTION 19

```
mysql> SELECT c.id,c.name,ci.address
-> FROM class c
-> INNER JOIN class_info ci
-> ON c.id=ci.id;
+-----+-----+-----+
| id  | name | address |
+-----+-----+-----+
| 1   | abhi | DELHI   |
| 2   | adam | MUMBAI  |
| 3   | alex | CHENNAI |
+-----+-----+-----+
3 rows in set (0.00 sec)
```

QUESTION 20

```
mysql> SELECT c.id,  
->         c.name,  
->         CASE  
->             WHEN ci.address IS NULL  
->             THEN 'Address Missing'  
->             ELSE 'Address Available'  
->         END AS Status  
-> FROM class c  
-> LEFT JOIN class_info ci  
-> ON c.id=ci.id;
```

id	name	Status
1	abhi	Address Available
2	adam	Address Available
3	alex	Address Available
4	anu	Address Missing
5	ashish	Address Missing

5 rows in set (0.00 sec)